QUANT

Ratio & Partnership, Mixtures & Alligations

Assignment 01

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Ratio Online Assignment -01

1. A piece of string 175 cm in length was cut into pieces, the ratio of whose lengths was 5 : 2. Find the length of shorter piece.

1. 55 cm 2.50 cm 3.40 cm 4. 45 cm

2. Which number should be added to each of the numbers 7, 16 and 34 so that the sums would be in continued proportion?

1. 2 2. 1 3.3 4. 4

3. Find a number such that when it is subtracted from each of 19, 26, 35 and 50, the remainders would be in proportion.

1. 5 2. 2 3.4 4. 6

4. On a map, the distance between A and B is 3 inch. The actual distance between these two is900 km. What is the actual distance from C to B, which are 4 inch apart on the map?

1. 1200 km 2.1250 km 3.1600 km 4. 1400 km

5. The time "*t*" to go over a fixed distance is in inverse proportion to the average speed "*v*" of the vehicle. When the average speed of the vehicle is 48 km/hr it takes 6 hours to go over a certain distance. Find the time required to cover the same distance at an average speed of 72 km/hr.

1. 3 hrs 2. 4 ½ hrs 3.4 hrs 4. 5 hrs

6. The mass of a piece of metal and its volume are in direct proportion to each other. The mass of a piece of metal having volume of 250 cm^3 is 1,950 gm. Find the volume of another piece of the same metal whose mass is 4,680 gm.

1. 370 cm^3 2. 600 cm^3 3.500 cm^3 4. 720 cm^3

7. The volume of a sphere varies directly as the cube of its radius. Three solid spheres of radii 4.5 cm, 6 cm and 7.5 cm were melted and cast into a big solid sphere. Find the radius of that sphere.

1. 9 cm 2..5 cm 3. 8 cm 4. 10 cm

8. The distance through which a heavy body falls down from rest varies directly as the square of time for which it is falling. In 5 seconds an object covers a distance of 78.4 metres more than the distance covered by it in 3 sec. Find the distance through which it will fall in 10 sec.

1. 640 m 2.800 m 3.180m 4. 490 m

9. The volume of a sphere and the cube of its diameter are in direct proportion. Two solid spheres of diameters 3 cm and 4 cm were melted together and cast into a hollow sphere of diameter 6 cm. Find the thickness of the outer cover of the sphere.

1. ½ cm 2. $^1/_3$ cm 3.¼ cm 4. 1 cm

10. The square of the time required by a planet to make a complete revolution around the sun varies as the cube of its mean distance from the sun. If the mean distance of Jupiter and the Earth from the sun are in the ratio 16 : 3, find the number of days on Earth in Jupiter's year.

1. 1,065 days 2.2,455 days 3.4,495 days 4. 3,675 days

Answer key & Explanations

Q No.	Key	Explanation
1.	2	Total length = 175 cm. Ratio is 5:2 $\therefore$ Length of each piece is $(175 \times 5) / 7 = 125$ cm and $(175 \times 2) / 7 = 50$ cm.
2.	1	Let the number to be added = x. $\therefore 7 + x/16 + x = 16 + x/34 + x$ $\Rightarrow x = 2$.
3.	1	Let the number to be subtracted = x. $\therefore 19 - x/26 - x = 35 - x/50 - x$ $\Rightarrow x = 5$.
4.	1	3 inch is the same as 900 km. $\therefore$ 1 inch = 300 km. Hence 4 inch = $4 \times 300 = 1{,}200$ km.
5.	3	$T \alpha \frac{1}{V} \Rightarrow T = \frac{k}{V} . \therefore 6 = \frac{k}{48}$ $\Rightarrow k = 48 \times 6$. Now $T = (48 \times 6)/72 = 4$ hrs.
6.	2	$M \alpha V$ $\Rightarrow M = kV$. $1950 = k . 250$ $\Rightarrow K = 195/25$. As $V = M/k$ $\therefore V = 4680 \times 25/195 = 600$ cm$^{3.}$
7.	1	$V \propto r^3$. Total volume = $(4.5)^3 + (6)^3 + (7.5)^3 = 729$ $\therefore$ required radius = $(729)^{1/3} = 9$ cm.
8.	4	$D \propto T^2$ $\Rightarrow D = kT^2$ $\Rightarrow k(5)^2 - k(3)^2 = 78.4$ $\Rightarrow 16k = 78.4$ $\Rightarrow k = 4.9$ $\therefore D = 4.9(10)^2 = 490$ m.
9.	1	$V \propto D^3$. Total volume of these 2 spheres = $(3)^3 + (4)^3$ $= 27 + 64 = 91$. Volume of the larger sphere = $(6)^3 = 216$ $\therefore$ Volume of the inner sphere = $216 - 91 = 125$. Hence radius of this sphere = $(125)^{1/3} = 5$. $\therefore$ Total thickness = $6 - 5 = 1$ cm. Hence thickness of outer cover = ½ = 0.5 cm.
10.	3	$T^2 \propto (MD)^3$ $\Rightarrow T^2 = k\,(MD)^3$. $MD_{\text{Jupiter}} : MD_{\text{Earth}} = 16:3$. Earth days = 365. Jupiter days =? $\therefore \left(\frac{J}{365}\right)^2 = \left(\frac{16}{3}\right)^3 \Rightarrow J^2 = 4096 \times 365 \times 365/27$ $\Rightarrow J = 4495$ days.